

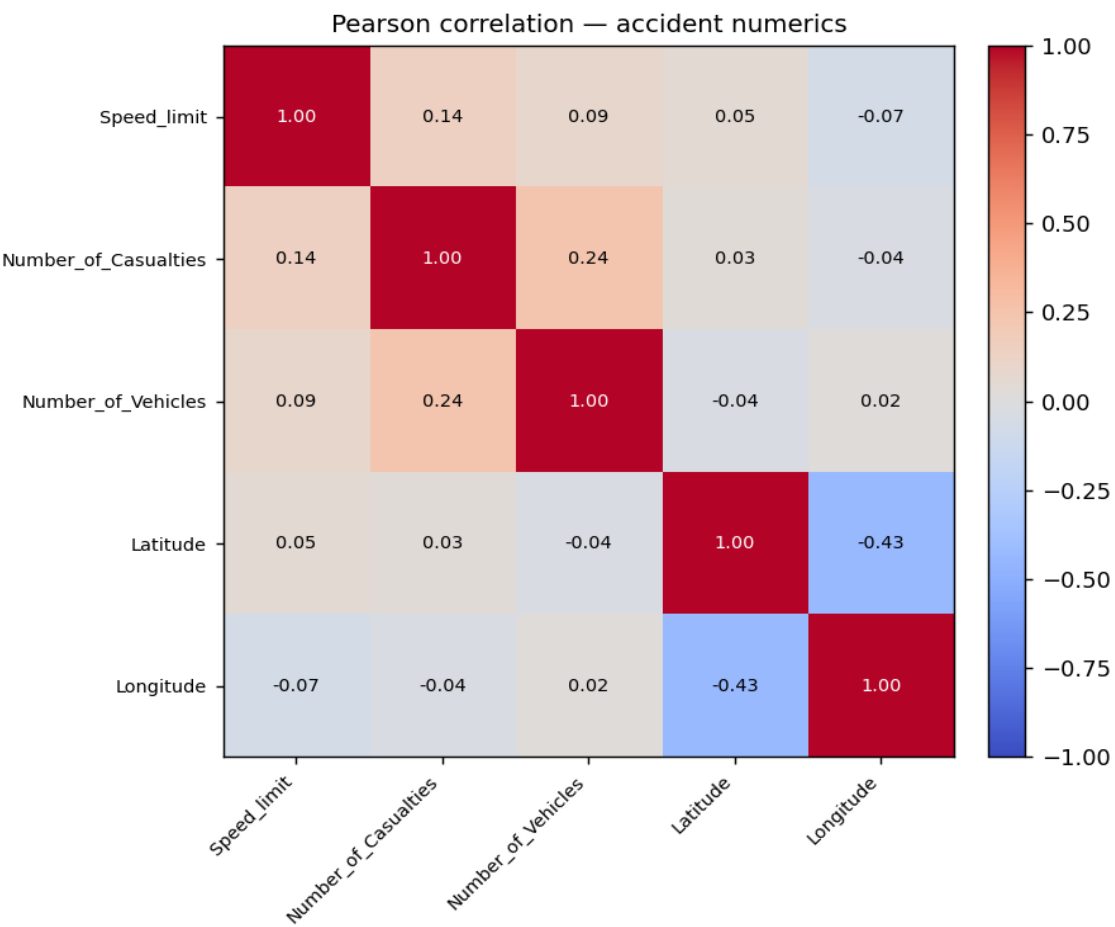
UK Road Accidents — Data Validation Report

Generated: 2026-05-02T05:35:17.191499

Total Issues Found: 18

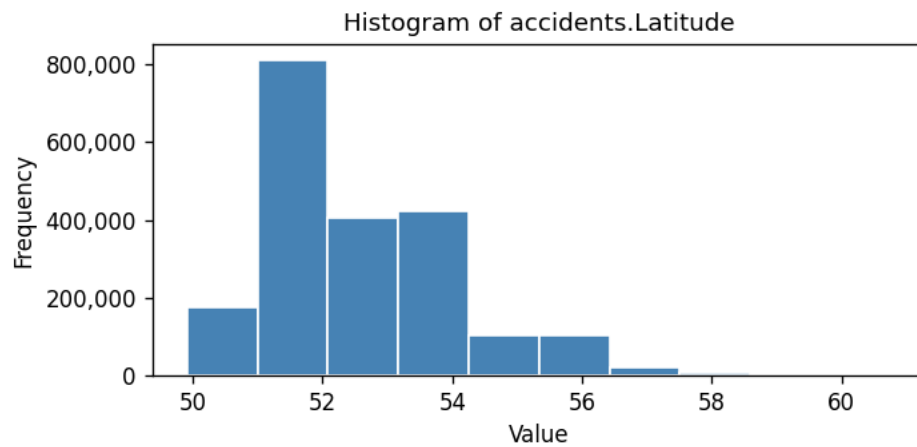
Dimension	Issues
ACCURACY	2
COMPLETENESS	3
JOIN_INTEGRITY	2
WEATHER_DISTRIBUTION	11

Correlation Heatmap

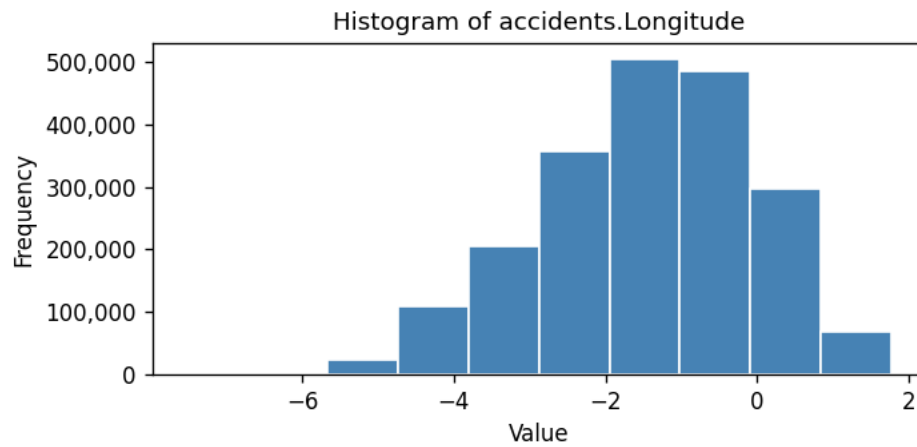


Distribution Histograms

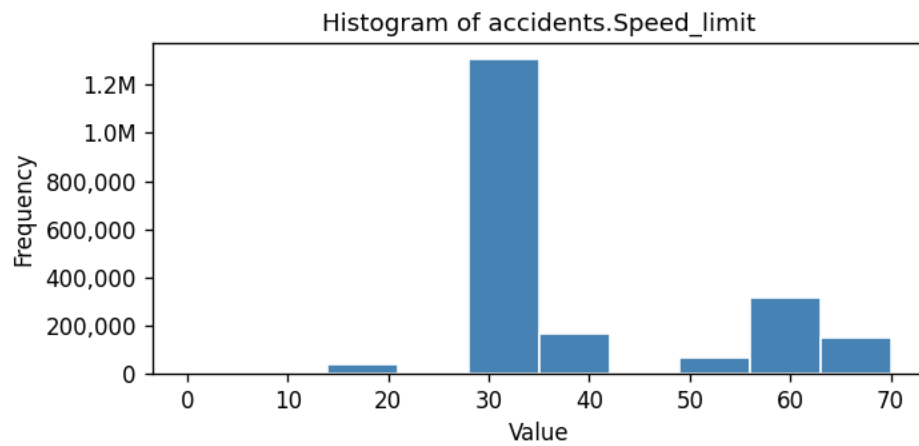
accidents.Latitude



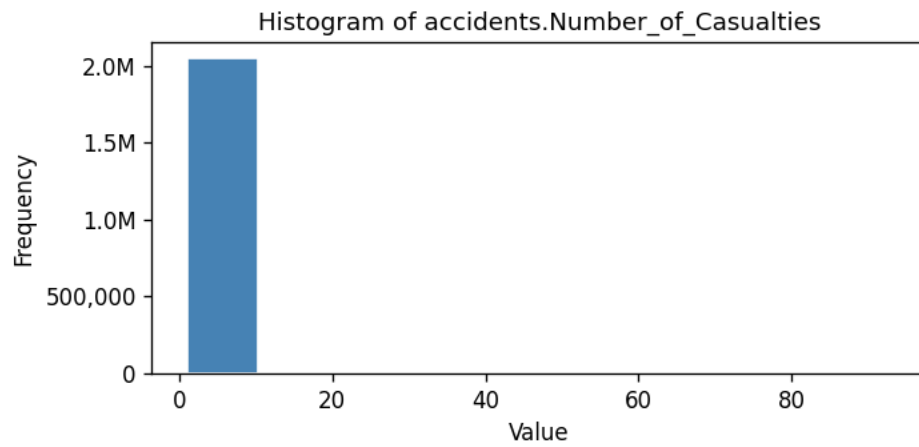
accidents.Longitude



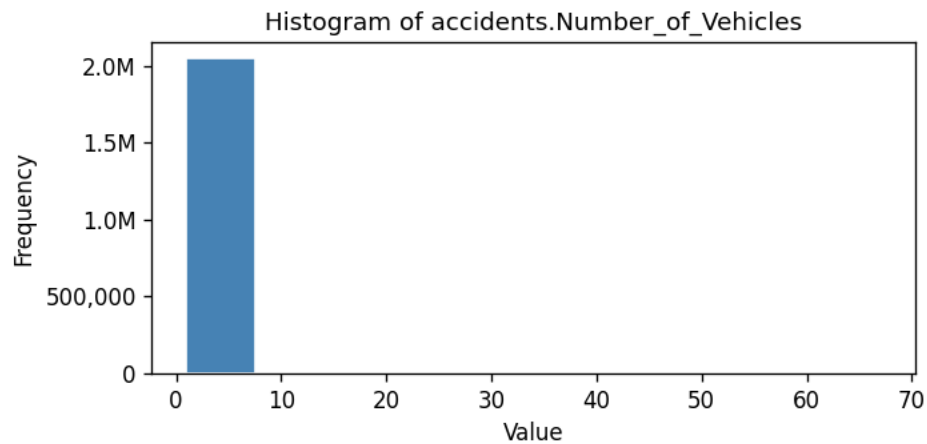
accidents.Speed_limit



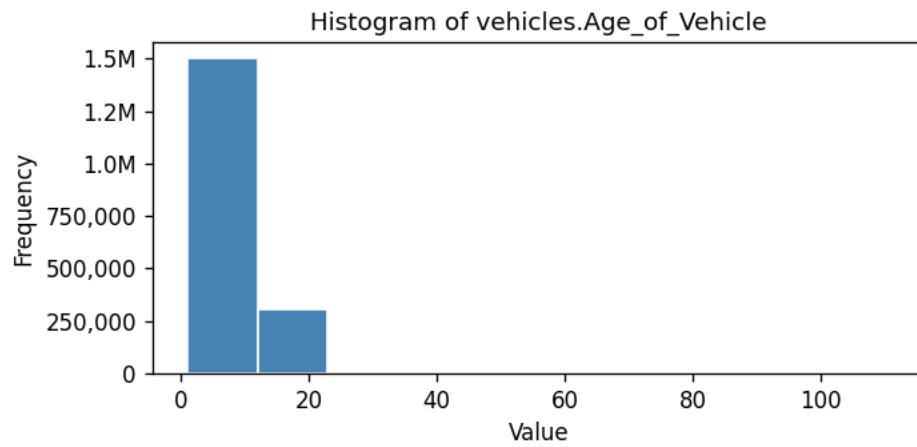
accidents.Number_of_Casualties



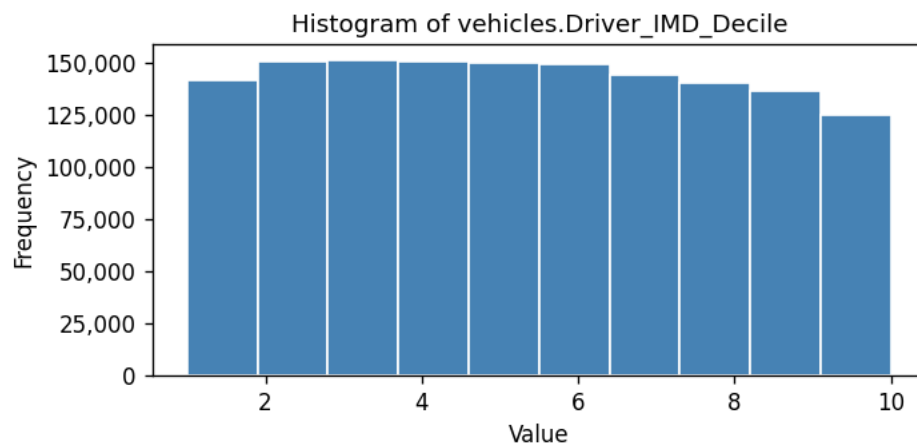
accidents.Number_of_Vehicles



vehicles.Age_of_Vehicle



vehicles.Driver_IMD_Decile



Full Validation Log

```
=====
UK Road Accidents – Data Validation Report
2026-05-02 05:35:20
=====

Loaded accidents : 2,047,256 rows x 34 columns
Loaded vehicles : 2,177,205 rows x 24 columns
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DIMENSION 1 – ACCURACY
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[PASS] Accident_Severity – all values valid (Slight/Serious/Fatal)
[ISSUE] 36 rows have Speed_limit not in UK legal set {50, 20, 70, 40, 60, 30}
[PASS] Latitude – all within UK bounds (49-61)
[PASS] Longitude – all within UK bounds (-8 to 2)
[PASS] Number_of_Vehicles – all values >= 1
[PASS] Age_of_Vehicle – no negative values
[INFO] Age_Band_of_Driver has 12 unique categories
Sample values: ['26 - 35', '66 - 75', '36 - 45', '46 - 55', '21 - 25', 'Data missing or out of range', '16 - 20', '
56 - 65', 'Over 75', '11 - 15']
[PASS] Day_of_Week – all values valid
[ISSUE] 76,119 vehicle records have invalid Sex_of_Driver values
-----

DIMENSION 2 – CONSISTENCY
-----

[PASS] Date column – parses correctly as datetime
[PASS] Year column – matches year extracted from Date in all rows
[WARN] 2 Date/Day_of_Week mismatches found
[PASS] Time column – all values follow HH:MM format
[PASS] Year – consistent across both tables for all matched accidents
-----

DIMENSION 3 – COMPLETENESS
-----

[ACCIDENTS]
Column Missing %
-----

Carriageway_Hazards 2,007,807 98.1%
Special_Conditions_at_Site 1,995,137 97.5%
2nd_Road_Class 844,272 41.2%
LSOA_of_Accident_Location 144,953 7.1%
2nd_Road_Number 17,593 0.9%
Pedestrian_Crossing-Physical_Facilities 3,560 0.2%
Pedestrian_Crossing-Human_Control 2,920 0.1%
Did_Police_Officer_Attend_Scene_of_Accident 278 0.0%
Latitude 174 0.0% [!] REQUIRED
Location_Easting_OSGR 164 0.0%
Location_Northing_OSGR 164 0.0%
Longitude 175 0.0% [!] REQUIRED
[ISSUE] [ACCIDENTS] CRITICAL: 'Latitude' has 174 nulls (0.01%). These rows must be dropped.
[ISSUE] [ACCIDENTS] CRITICAL: 'Longitude' has 175 nulls (0.01%). These rows must be dropped.
[ISSUE] [ACCIDENTS] CRITICAL: 'Time' has 156 nulls (0.01%). These rows must be dropped.
[ADVICE] 'Carriageway_Hazards' is 98.07% missing. Consider dropping this feature.
[ADVICE] 'Special_Conditions_at_Site' is 97.45% missing. Consider dropping this feature.
[VEHICLES]
Column Missing %
-----

Hit_Object_in_Carriageway 2,087,824 95.9%
Hit_Object_off_Carriageway 1,989,741 91.4%
Skidding_and_Overturning 1,898,208 87.2%
Driver_IMD_Decile 734,812 33.8%
Age_of_Vehicle 358,149 16.4%
model 325,331 14.9%
Engine_Capacity_.CC. 265,861 12.2%
Propulsion_Code 245,843 11.3%
make 110,845 5.1%
Vehicle_Location.Restricted_Lane 1,317 0.1%
[PASS] All strictly required columns in VEHICLES are 100% complete.
[ADVICE] 'Hit_Object_in_Carriageway' is 95.89% missing. Consider dropping this feature.
```



```

[ 7.35 - 11.46] 351,669 ( 25.3%) ■■■■■■■■■■■■
[ 11.46 - 15.57] 383,705 ( 27.6%) ■■■■■■■■■■■■
[ 15.57 - 19.68] 226,063 ( 16.3%) ■■■■■■■■■■
[ 19.68 - 23.79] 36,972 ( 2.7%) ■
[ 23.79 - 27.90] 3,274 ( 0.2%) ■
KS=0.0464 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'tmin' is 56.8% missing (>20.0% threshold)
tmin 56.8% -24.20 26.10 7.12 7.40 5.20 -0.24 -0.51
[PASS] tmin – all values within physical bounds [-30, 45]
Histogram (tmin):
[ -24.20 - -19.17] 15 ( 0.0%) ■
[ -19.17 - -14.14] 79 ( 0.0%) ■
[ -14.14 - -9.11] 879 ( 0.1%) ■
[ -9.11 - -4.08] 14,275 ( 1.2%) ■
[ -4.08 - 0.95] 137,593 ( 11.7%) ■■■■■■
[ 0.95 - 5.98] 307,622 ( 26.2%) ■■■■■■■■■■■■
[ 5.98 - 11.01] 422,566 ( 36.0%) ■■■■■■■■■■■■
[ 11.01 - 16.04] 268,183 ( 22.9%) ■■■■■■■■■■■■
[ 16.04 - 21.07] 22,117 ( 1.9%) ■
[ 21.07 - 26.10] 190 ( 0.0%) ■
KS=0.0585 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'tmax' is 56.7% missing (>20.0% threshold)
tmax 56.7% -8.00 36.70 14.22 14.10 5.88 0.04 -0.40
[PASS] tmax – all values within physical bounds [-25, 50]
Histogram (tmax):
[ -8.00 - -3.53] 252 ( 0.0%) ■
[ -3.53 - 0.94] 5,819 ( 0.5%) ■
[ 0.94 - 5.41] 75,094 ( 6.4%) ■■■■
[ 5.41 - 9.88] 201,948 ( 17.2%) ■■■■■■■■
[ 9.88 - 14.35] 314,318 ( 26.7%) ■■■■■■■■■■■■
[ 14.35 - 18.82] 299,756 ( 25.5%) ■■■■■■■■■■■■
[ 18.82 - 23.29] 211,476 ( 18.0%) ■■■■■■■■■■
[ 23.29 - 27.76] 54,974 ( 4.7%) ■■
[ 27.76 - 32.23] 11,470 ( 1.0%) ■
[ 32.23 - 36.70] 603 ( 0.1%) ■
KS=0.0460 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'prcp' is 94.5% missing (>20.0% threshold)
prcp 94.5% 0.00 221.50 2.01 0.30 4.44 6.48 147.83
[PASS] prcp – all values within physical bounds [0, 10000]
Histogram (prcp):
[ 0.00 - 22.15] 148,338 ( 99.2%) ■■■■■■■■■■■■
[ 22.15 - 44.30] 1,050 ( 0.7%) ■
[ 44.30 - 66.45] 78 ( 0.1%) ■
[ 66.45 - 88.60] 15 ( 0.0%) ■
[ 88.60 - 110.75] 0 ( 0.0%) ■
[ 110.75 - 132.90] 0 ( 0.0%) ■
[ 132.90 - 155.05] 0 ( 0.0%) ■
[ 155.05 - 177.20] 0 ( 0.0%) ■
[ 177.20 - 199.35] 0 ( 0.0%) ■
[ 199.35 - 221.50] 3 ( 0.0%) ■
KS=0.3245 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'snwd' is 98.7% missing (>20.0% threshold)
snwd 98.7% 0.00 133.00 0.61 0.00 6.75 17.12 306.90
[PASS] snwd – all values within physical bounds [0, 2000]
Histogram (snwd):
[ 0.00 - 13.30] 34,789 ( 99.4%) ■■■■■■■■■■■■
[ 13.30 - 26.60] 74 ( 0.2%) ■
[ 26.60 - 39.90] 0 ( 0.0%) ■
[ 39.90 - 53.20] 0 ( 0.0%) ■
[ 53.20 - 66.50] 12 ( 0.0%) ■
[ 66.50 - 79.80] 8 ( 0.0%) ■
[ 79.80 - 93.10] 15 ( 0.0%) ■
[ 93.10 - 106.40] 5 ( 0.0%) ■
[ 106.40 - 119.70] 23 ( 0.1%) ■
[ 119.70 - 133.00] 57 ( 0.2%) ■
KS=0.4684 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'wspd' is 51.7% missing (>20.0% threshold)
wspd 51.7% 0.00 90.90 15.54 14.40 7.25 1.24 3.82
[PASS] wspd – all values within physical bounds [0, 114]
Histogram (wspd):

```

```
KS=0.0735 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'pres' is 73.9% missing (>20.0% threshold)
pres 73.9% 955.20 1044.20 1014.84 1015.70 10.78 -0.46 0.43
[PASS] pres – all values within physical bounds [870, 1085]
```

Histogram (pres):

[illegible]

```
KS=0.0416 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'rhum' is 49.4% missing (>20.0% threshold)
rhum 49.4% 4.00 100.00 79.99 81.00 9.81 -0.44 -0.10
[PASS] rhum – all values within physical bounds [0, 100]
```

Histogram (rhum):

[illegible]

```
KS=0.0652 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'wpgt' is 100.0% missing (>20.0% threshold)
wpgt 100.0% [SKIP – too few valid values]
[ISSUE] [WEATHER] 'tsun' is 100.0% missing (>20.0% threshold)
tsun 100.0% [SKIP – too few valid values]
[ISSUE] [WEATHER] 'cldc' is 85.6% missing (>20.0% threshold)
cldc 85.6% 0.00 8.00 5.09 5.00 1.80 -0.71 0.23
[PASS] cldc – all values within physical bounds [0, 9]
```

Histogram (cldc):

Interval	Count	Percentage	Bar
[0.00 - 0.80]	9,133	(2.3%)	■
[0.80 - 1.60]	8,549	(2.2%)	■
[1.60 - 2.40]	18,176	(4.6%)	■■
[2.40 - 3.20]	29,460	(7.5%)	■■■
[3.20 - 4.00]	0	(0.0%)	■
[4.00 - 4.80]	68,104	(17.4%)	■■■■■■■■■
[4.80 - 5.60]	71,745	(18.3%)	■■■■■■■■■■
[5.60 - 6.40]	96,445	(24.6%)	■■■■■■■■■■■■
[6.40 - 7.20]	69,912	(17.9%)	■■■■■■■■■
[7.20 - 8.00]	19,925	(5.1%)	■■■

KS=0.1647 p=0.000000 → NOT normal (p<0.05) – use Spearman

DIMENSION 8 – RELATIONSHIPS

Pearson Correlation Matrix:

```
Speed_limit Number_of_C Number_of_V Latitude Longitude
Speed_limit 1.000 0.143 0.088 0.052 -0.066
Number_of_Casualties 0.143 1.000 0.237 0.030 -0.038
Number_of_Vehicles 0.088 0.237 1.000 -0.039 0.019
Latitude 0.052 0.030 -0.039 1.000 -0.434
Longitude -0.066 -0.038 0.019 -0.434 1.000
```

Pearson vs Spearman (|diff| > 0.10 = non-linear relationship detected):
Pair Pearson Spearman |Diff| Note

Speed_limit x Number_of_Casualties 0.143 0.162 0.019
Speed_limit x Number_of_Vehicles 0.088 0.069 0.019
Speed_limit x Latitude 0.052 0.034 0.018
Speed_limit x Longitude -0.066 -0.062 0.004
Number_of_Casualties x Number_of_Vehicles 0.237 0.230 0.007
Number_of_Casualties x Latitude 0.030 0.044 0.014
Number_of_Casualties x Longitude -0.038 -0.046 0.008
Number_of_Vehicles x Latitude -0.039 -0.029 0.010
Number_of_Vehicles x Longitude 0.019 0.018 0.001
Latitude x Longitude -0.434 -0.411 0.023
[PASS] No high-correlation pairs found (threshold |r| > 0.85)
[SAVED] reports\correlation_heatmap.png

JOIN INTEGRITY (Accident_Index key across both tables)

Unique Accident_Index in accidents table : 2,047,256
Unique Accident_Index in vehicles table : 1,488,981
Keys matched (will survive inner join) : 1,389,724
[ISSUE] 657,532 accidents (32.1%) have no vehicle record – dropped on inner join, document in cleaning step
[ISSUE] 99,257 vehicle records (6.7%) have no matching accident – orphaned records, dropped on inner join
=====

VALIDATION COMPLETE

Total issues found across all 8 dimensions: 18

Issues by dimension:

[ACCURACY] 36 rows have Speed_limit not in UK legal set {50, 20, 70, 40, 60, 30}
[ACCURACY] 76,119 vehicle records have invalid Sex_of_Driver values
[COMPLETENESS] [ACCIDENTS] CRITICAL: 'Latitude' has 174 nulls (0.01%). These rows must be dropped.
[COMPLETENESS] [ACCIDENTS] CRITICAL: 'Longitude' has 175 nulls (0.01%). These rows must be dropped.
[COMPLETENESS] [ACCIDENTS] CRITICAL: 'Time' has 156 nulls (0.01%). These rows must be dropped.
[WEATHER_DISTRIBUTION] [WEATHER] 'temp' is 48.8% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'tmin' is 56.8% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'tmax' is 56.7% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'prcp' is 94.5% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'snwd' is 98.7% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'wspd' is 51.7% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'pres' is 73.9% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'rhum' is 49.4% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'wpgt' is 100.0% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'tsun' is 100.0% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'cldc' is 85.6% missing (>20.0% threshold)
[JOIN_INTEGRITY] 657,532 accidents (32.1%) have no vehicle record – dropped on inner join, document in cleaning step
[JOIN_INTEGRITY] 99,257 vehicle records (6.7%) have no matching accident – orphaned records, dropped on inner join
[SAVED] reports\validation_report.json
[SAVED] reports\validation_summary.txt